NetShield: Feasibility Boundaries and Shared-Bottleneck Evidence Admission for Multi-Probe IoT Edge Intrusion Detection

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Abstract—Multi-probe edge intrusion detection creates an evidence-service allocation problem: concurrent probes may request evidence whose aggregate demand exceeds a shared service budget. We develop NetShield around an analytical feasibility region rather than classifier-accuracy optimization. For n probes, let m probes retain remotely delivered evidence in a coordination round, of which k request full evidence of size B while the remaining $m-k$ transmit summaries of size $S < B$. The joint request is byte-feasible within one idealized service round if and only if $kB + (m-k)S \le C$. For a fixed remote-evidence coverage requirement m, this gives the exact full-evidence boundary $k^\star = \min\{m, \lfloor (C - mS)/(B - S) \rfloor\}$ when $C \ge mS$, exposing an explicit coverage–richness trade-off.

We evaluate the resulting design on a heterogeneous Raspberry Pi 5, Pi 4B, and Pi 3B+ testbed over Wi-Fi. With full and summary payloads of 128 KiB and 16 KiB, respectively, the low/mid/high formal service budgets predict exactly 0/1/3 simultaneous full-evidence actions when all three probes retain remote evidence; these operating points match the coordinated action mixes in every formal session. We further identify a symmetry-induced limit cycle in the frozen independent queue-penalized controller. At mid capacity, a synchronized full-evidence round leaves approximately 43.7 KiB of backlog per probe, while every positive-payload action loses to the best zero-payload action once queue occupancy exceeds approximately 260 B. This produces the observed full/none period-two orbit.

A 12-value offline sensitivity sweep exactly reproduces the formal $v = 1$ operating statistics and shows that the synchronized mid-capacity core-request fraction remains 1/2 through $v = 30{,}000$. A separate 20-seed software sensitivity study breaks only the probe phases; the desynchronized independent controller reaches a 2/3 core-request fraction and approximately 0.50 simulated on-time coverage, showing that the original baseline failure is caused by synchronization rather than decentralization itself. We therefore interpret coordinated admission as deterministic selection of a coverage–richness operating point, not as universally superior control.

The physical campaign contains 12 sessions, 216 accepted arms, 648 device-runs, and 58,320 measured events. A timestamp reconstruction separates a 1-s post-release service SLO from schedule-inclusive latency, and a discrete-event conformance study matches the analytical integer boundary for all 20 tested probe-count/capacity combinations. A full-raw audit reconstructs all 216 accepted hardware arms with zero raw-to-metric mismatch and zero shared-cap violation across 63,720 service rounds. These results position edge-IDS evidence scheduling as a shared-bottleneck feasibility and service-timeliness problem rather than a classifier-performance optimization problem.

Index Terms—Internet of Things, intrusion detection, edge computing, admission control, shared bottleneck, evidence scheduling, latency SLO, queue-aware control.

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I. Introduction

Intrusion detection is increasingly moved toward gateways and edge nodes to reduce cloud dependence, shorten response paths, and keep sensitive telemetry close to its source. In a multi-probe deployment, however, a security decision produces more than a class label. A probe may retain a result locally, send a compact alert or summary, or upload richer evidence for core-side execution and correlation. These alternatives create different amounts of network demand and different completion paths.

This coupling changes the systems problem. Suppose three probes each decide independently that a 128 KiB full-evidence request is useful. Every local decision may be reasonable in isolation, yet their combined 393,216 B request can exceed the shared service available in the current coordination round. The resulting question is not merely whether each probe can classify its traffic, but which combination of evidence coverage and evidence richness can be admitted over the common bottleneck.

This paper studies that question through NetShield. The key modeling choice is to treat evidence generation as endogenous application load: the runtime action determines how many bytes enter the service plane. This differs from treating security traffic as a fixed exogenous arrival process. It also separates two mechanisms that are easily conflated. A selector determines which action a probe would like to take; a joint admission layer determines whether the simultaneous requests form a feasible tuple before those bytes enter the queues.

The distinction became especially important during audit of our independent queue-aware baseline. The frozen formal controller used the score

$$\psi_i(a,t) = v u(a) - Q_i(t) b(a)/10^6, \tag{1}$$

where Q_i is the local evidence queue and $b(a)$ the payload of action a. Because the three formal probes execute synchronized decision epochs, their local queue states remain symmetric. Under the frozen $v = 1$ scaling, this symmetry creates a deterministic period-two orbit at mid capacity: all probes request full evidence together, retain residual queue, then all switch to a zero-payload action together until the queue drains. Consequently, the poor mid-capacity result of this baseline cannot be used as evidence that decentralization is intrinsically inferior.

Rather than hiding this behavior, V1.3 makes it a first-class systems result. We derive a closed-form evidence feasibility region, show how the synchronized local controller can cycle relative to that region, and then use both a desynchronized counterfactual and the physical campaign to distinguish three phenomena: deterministic byte feasibility, controller-induced operating-point selection, and real-system timing variation.

We address four research questions:

- RQ1—Feasibility region: What coverage–richness evidence allocations are byte-feasible under a shared service budget?
- RQ2—Controller realization and synchronization: How do coordinated and independent controllers occupy that region, and when does synchronized local queue feedback create limit cycles?
- RQ3—Physical timing residual: Once an evidence tuple is byte-feasible, how do orchestration and wireless timing affect post-release and schedule-inclusive service attainment?
- RQ4—Scaling and reproducibility: Does the finite-probe analytical boundary hold as the number of probes changes, and can the physical result chain be reconstructed from raw occurrence-level evidence?

The paper makes four contributions.

First, we derive a coverage-conditioned evidence feasibility boundary. For a shared budget C, full payload B, summary payload S, m remotely represented probes, and k full-evidence probes, feasibility is exactly characterized by $kB+(m-k)S \le C$. Its normalized form yields a simple coverage–richness frontier.

Second, we identify and analytically explain a symmetry-induced limit cycle in synchronized independent queue-penalized control. A 12-value parameter sweep reproduces the formal operating point over more than four orders of magnitude in v, while a 20-seed desynchronization study shows that removing the shared phase substantially changes the independent operating point without introducing joint admission.

Third, we validate the analytical feasibility and overload boundaries on three heterogeneous physical Raspberry Pi probes. The formal campaign contains 12 sessions, 216 accepted arms, and 58,320 measured events. The hardware campaign is used to characterize the deviations that the byte model cannot predict, in particular evidence completion latency and release-time jitter.

Fourth, we provide an event-level reproducibility chain. The accepted hardware arms are reconstructed from immutable event contracts, and a separate methodology-repair package validates the frozen controller, the desynchronization counterfactual, schedule-inclusive timing, and multi-n boundary conformance.

a) Claim boundary.: NetShield does not claim a new intrusion classifier, state-of-the-art CICIoT2023 detection accuracy, global scheduler optimality, or empirical superiority of the safety filter over ordinary coordinated admission. The central claims concern evidence feasibility, admission, queue behavior, and service timing.

II. Related Work

A. Lightweight IoT Intrusion Detection

Lightweight IoT IDS research has produced compact anomaly detectors, device fingerprints, and online detection pipelines, including SVELTE, IoT Sentinel, N-BaIoT, and Kitsune [1]–[4]. These systems primarily ask how to make prediction feasible on constrained devices. NetShield addresses a complementary question: given a frozen detector/evidence action set, which evidence requests can be sustained jointly over a shared edge service?

B. Edge Offloading and Shared Network Control

Mobile-edge computing studies computation placement under latency, bandwidth, and energy constraints [5], [6]. Classical fair queueing and network scheduling address how a bottleneck serves packets that have already entered queues [7], [8]. Our setting adds an upstream control decision: selecting an evidence action determines the amount of new traffic offered to the service plane. Admission and service are therefore modeled separately.

C. Queue-Aware Control

Queue-aware scheduling and Lyapunov drift-plus-penalty provide a foundation for balancing utility against backlog [9], [10]. The frozen experiment artifacts historically use the labels “independent-DPP,” “central-DPP,” and “shielded-DPP.” The deployed selector, however, is the finite-action score in (1), not a claimed implementation of a complete stochastic Lyapunov optimizer. V1.3 therefore uses the descriptive names independent queue-penalized greedy (I-QPG), coordinated QPG (C-QPG), and safety-filtered coordinated QPG (S-CQPG), while retaining the old identifiers only for artifact traceability.

D. Deadline-Aware Inference Serving and Admission

Prediction-serving systems also expose tight latency-resource trade-offs. Clipper combines batching, caching, and adaptive model selection for low-latency prediction serving [11]. Nexus schedules DNN execution across GPU clusters while targeting latency constraints [12]. Clockwork constructs predictable DNN serving from predictable execution primitives and explicitly targets request-level latency objectives [13]. InferLine combines offline provisioning with online tuning to meet end-to-end tail-latency SLOs for prediction pipelines [14].

These systems primarily control compute allocation, batching, provisioning, or execution after inference requests exist. NetShield instead controls an upstream networking decision in which the security action itself determines the amount and richness of evidence offered to the shared transport-and-core service.

E. Runtime Safety Filters

Runtime shields and predictive safety filters separate policy choice from hard enforcement [15], [16]. We retain that architecture as an optional local-feasibility filter. The formal campaign shows that the filter is active under low capacity, but also that ordinary capacity admission can produce the same final action mix. We therefore do not treat the filter as an independently validated performance contribution.

F. Relation to Prior Work

Our prior ACISP 2026 Systematization of Knowledge surveyed telemetry-aware runtime assurance for always-on on-device IDS and presented TQS-IDS as a design template [17]. It did not implement the present shared-bottleneck formulation, the queue-penalized controllers, the V1.2 event contract, or the S01–S12 three-Pi hardware campaign.

III. System Model and Feasibility Analysis

A. Shared Evidence Service

Consider n edge probes sharing an application-level service budget $C(t)$ bytes in coordination round t. Probe i has evidence queue $Q_i(t)$ and chooses an admitted action $a_i(t)$ with payload $b(a_i)$. The common service grants $g_i(t)$ satisfy

$$g_i(t) \geq 0, \qquad \sum_{i=1}^{n} g_i(t) \leq C(t). \tag{2}$$

The queue evolves as

$$Q_i(t+1) = \left[Q_i(t) + b(a_i(t)) - g_i(t)\right]^+ . \tag{3}$$

The low, mid, and high labels used in the formal campaign refer to controlled application-byte budgets per coordination round. They are not measurements of Wi-Fi PHY throughput.

B. Frozen Action Lattice

The implementation exposes ten frozen actions spanning light, medium, and heavy detector execution and evidence levels from no upload to full core-side evidence. Table I lists the frozen payload and utility fields.

The formal coordinated regimes eventually operate on the H-Summary–H-Full edge of this lattice, making it possible to state a compact exact boundary.

TABLE I
Frozen action payloads and utility scores.

Action	Payload (B)	Utility
L-NONE	0	0.758501
L-ALERT	256	0.825168
L-SUMMARY	4,096	0.891835
M-NONE	0	0.766968
M-ALERT	256	0.833635
M-SUMMARY	8,192	0.900302
H-NONE	0	0.768378
H-ALERT	256	0.835044
H-SUMMARY	16,384	0.901711
H-FULL	131,072	0.968378

C. Coverage–Richness Feasibility Region

Let $m \leq n$ probes retain remotely delivered evidence in a round. Among them, let $k \leq m$ request full evidence of size B; the other $m-k$ send summaries of size S, where $0 < S < B$. The remaining $n-m$ probes do not generate remote evidence.

Theorem 1 (Coverage-conditioned evidence feasibility). A (k, m) allocation is byte-feasible within one idealized service round if and only if

$$kB + (m-k)S \leq C. \tag{4}$$

For a fixed remote-evidence requirement m, if $C \geq mS$, the maximum feasible number of full-evidence probes is

$$k^\star(C, m) = \min\left\{m, \left\lfloor \frac{C - mS}{B - S} \right\rfloor\right\}. \tag{5}$$

If $C < mS$, no allocation preserving remote evidence for all m probes is feasible.

Proof. The payload of a (k, m) allocation is

$$kB + (m-k)S = mS + k(B-S).$$

Since $B - S > 0$, the capacity condition is equivalent to $k(B-S) \leq C - mS$. The largest integer satisfying this inequality is $\lfloor (C - mS)/(B - S) \rfloor$, subject to $0 \leq k \leq m$. □

Define normalized capacity, summary ratio, remote-evidence coverage, and full-evidence richness as

$$\rho = \frac{C}{nB}, \qquad \sigma = \frac{S}{B}, \qquad r = \frac{m}{n}, \qquad f = \frac{k}{n}. \tag{6}$$

Ignoring finite-n integer granularity, (4) becomes

$$\boxed{\sigma r + (1-\sigma) f \leq \rho} \tag{7}$$

or

$$f \leq \frac{\rho - \sigma r}{1 - \sigma}. \tag{8}$$

Equation (7) is the central service frontier of the paper. Increasing remote-evidence coverage r consumes budget even when that coverage is only summary-level; increasing full-evidence richness f consumes the remaining budget at slope $(1 - \sigma)$.

a) Summary-preserving special case.: When all probes retain remote evidence, $m = n$. Then

$$k^{\star}(C, n) = \left[\left\lfloor \frac{C - nS}{B - S} \right\rfloor\right]_0^n. \tag{9}$$

This is the exact boundary implemented by the formal coordinated full-to-summary degradation pattern.

D. Deterministic Overload Growth

The feasibility boundary also predicts queue growth for an unadmitted fixed-full stream.

Proposition 1 (Fixed-full overload law). Assume every probe generates B bytes per generation round, the initial queues are empty, the aggregate service is work conserving, and $C < nB$. After T generation rounds the aggregate backlog is

$$Q_{\Sigma}(T) = T(nB - C). \tag{10}$$

Under symmetric fair service, per-probe backlog is approximately

$$Q_i(T) = T\left(B - \frac{C}{n}\right), \tag{11}$$

up to integer-grant rounding.

Proof. Each round adds exactly nB new bytes and can remove at most C. Because $nB > C$, the queue never empties after overload begins, so the net backlog increase is exactly $nB - C$ per round. □

E. Independent Queue-Penalized Selection

The frozen independent policy uses

$$\psi_i(a, t) = vu(a) - Q_i(t)b(a)/10^6. \tag{12}$$

At $v = 1$, a positive-payload action a can beat H-None only while

$$Q_i < q_a = \frac{[u(a) - u(\text{H-None})]10^6}{b(a)}. \tag{13}$$

For the relevant actions,

$$q_{\text{H-Full}} \approx 1.526 \text{ B}, \quad q_{\text{H-Summary}} \approx 8.138 \text{ B},$$

and the largest threshold over any positive-payload frozen action is approximately 260.4 B, obtained by H-ALERT.

At mid capacity, a synchronized three-probe full-evidence request leaves approximately

$$q^{+} = B - \frac{2B}{3} = \frac{B}{3} \approx 43.7 \text{ KiB} \tag{14}$$

per probe after one service round.

Proposition 2 (Symmetry-induced period-two orbit). Suppose synchronized probes have identical selector state and empty queues, choose H-Full when $Q_i = 0$, and a synchronized H-Full round leaves residual queue q^{+}. If

$$q^{+} > \max_{a:b(a)>0} q_a \tag{15}$$

and a zero-payload round provides enough service to drain q^{+}, then the synchronized controller enters the period-two orbit

$$\text{H-Full} \rightarrow \text{H-None} \rightarrow \text{H-Full} \rightarrow \cdots. \tag{16}$$

Proof. After the synchronized H-Full round, $Q_i = q^{+}$ exceeds every threshold in (13); hence every positive-payload action scores below H-None. With no new payload, the next service round drains q^{+}. The state returns to $Q_i = 0$, at which point H-Full again maximizes the frozen utility score. Symmetry makes all probes repeat the same sequence. □

This proposition is a diagnosis of the frozen independent controller, not a statement that decentralized control must oscillate.

F. Joint Admission and Optional Safety Filtering

C-QPG first obtains per-probe queue-penalized requests and then monotonically reduces simultaneous requests until the common budget is satisfied. S-CQPG adds a local safety filter before that same joint admission stage. All policies use the same downstream shared service.

The safety filter is intentionally not part of Theorem 1. The theorem characterizes the network byte region; a local filter can additionally reject an action for device-specific constraints. In the present formal campaign, the low-capacity safety predicate and the shared-capacity admission stage ultimately select the same final summary mix, so the experiment does not establish an independent performance advantage for the filter.

IV. Experimental Methodology

A. Physical Testbed

The formal testbed uses three physical probes: P1 is a Raspberry Pi 5, P2 a Raspberry Pi 4B, and P3 a Raspberry Pi 3B+. Probe-to-core experiment traffic uses Wi-Fi. The edge core executes the heavy core path for H-Full events.

The workload is a frozen CICIoT2023-derived feature-vector replay rather than packet replay. Detector models, feature order, action profiles, window plans, and formal configuration are frozen before the campaign.

Each arm uses 5 warm-up rounds followed by 90 measured event occurrences per probe, with up to 200 drain rounds. The paper-eligible matrix is

$$12 \text{ sessions} \times 6 \text{ methods} \times 3 \text{ capacities} = 216 \text{ arms},$$

giving 648 device-runs and

$$648 \times 90 = 58,320$$

measured events.

B. Shared Service Budgets

The formal application-byte budgets are shown in Table II. Because $B = 128$ KiB, they correspond to one, two, or three full-evidence equivalents.

These budgets regulate application payload service. They must not be interpreted as measured physical 1/2/3-Mbit/s Wi-Fi links.

TABLE II
Formal application-service budgets.

Regime	Budget (B/round)	Full equivalents
Low	131,072	1
Mid	262,144	2
High	393,216	3

C. Methods

D. Event-Level Metric Authority

The occurrence-level event ledger is the scientific source of truth. Each admitted event retains an immutable execution contract specifying whether its result is local or core-dependent and what evidence transfer is required.

For a core-required event e, post-release latency is

$$L_{\text{post}}(e) = t_{\text{return}}(e) - t_{\text{created}}(e). \tag{17}$$

The primary timing SLO is

$$L_{\text{post}} \leq 1 \text{ s}. \tag{18}$$

The schedule-inclusive sensitivity instead uses

$$L_{\text{nom}}(e) = t_{\text{return}}(e) - t_{\text{scheduled}}(e). \tag{19}$$

This distinction is necessary because physical probes can release an event later than its nominal coordination-round time.

The primary service quantities are: (i) core-required fraction; (ii) core-result return coverage; (iii) post-release on-time core-result coverage; (iv) conditional post-release on-time rate among core-required events; (v) generated application payload; and (vi) queue occupancy.

The historical “timely-F1” field is not used. In the formal replay, classification correctness is constant for the scored occurrences, so adding “correct” to the main service metric does not provide additional information.

E. Statistical Reporting

We distinguish deterministic and timing-derived quantities. Action composition, core-required fractions, exact byte totals, analytical queue growth, and admission counts are deterministic under a frozen session plan and are reported exactly without confidence intervals.

Timing-derived quantities vary across the physical sessions. For these we treat the formal session as the independent replicate ($n = 12$) and report session-level means and descriptive session-bootstrap intervals where a paired contrast is useful. These intervals are descriptive; V1.3 does not carry forward the superseded non-inferiority analysis.

F. Methodology-Repair Sensitivities

Four post-audit analyses were run without altering the accepted hardware evidence.

A. v sensitivity. The exact frozen I-QPG objective, action order, and integer max-min allocator were replayed over 12 formal seeds for

$$v \in \{1, 3, 10, 30, 100, 168, 300, 1000, 3000, 10000, 30000, 100000\}. \tag{20}$$

The $v = 1$ replay is required to reproduce the formal D action mix, payload, core fraction, and sampled maximum queue before the sweep is accepted.

B. Desynchronization sensitivity. Twenty fixed seeds assign each probe an independent initial phase in $[0, 1)$ s. Each probe observes only its own queue and exchanges no action or queue state; no joint admission is introduced. This experiment is a discrete-event software sensitivity, not a hardware measurement.

C. Nominal-anchor reconstruction. Both (17) and (19) are reconstructed directly from accepted hardware timestamps for all 648 session/method/capacity/ device cells.

D. Probe-count conformance. A discrete-event simulator evaluates $n \in \{3, 8, 16, 32\}$ and $\rho \in \{1/3, 1/2, 2/3, 3/4, 1\}$ for the summary-preserving $m = n$ case. Its sole purpose is to test conformance with the integer boundary (9); it is not presented as a 32-node physical performance experiment.

V. Results

A. RQ1: The Analytical Boundary Predicts the Formal Coordinated Operating Points

For the formal coordinated testbed,

$$n = 3, \quad B = 128 \text{ KiB}, \quad S = 16 \text{ KiB},$$

so $\sigma = S/B = 1/8$.

When all probes retain remote evidence ($m = n = 3$), Theorem 1 predicts the operating points in Table IV. The formal C-QPG and S-CQPG action mixes match all three predictions in every session.

This result also clarifies what coordinated admission optimizes in the formal controller. At mid capacity, keeping all three probes represented remotely limits full evidence to one probe per idealized synchronous round. If the system relaxes the remote-evidence coverage requirement to $m = 2$, the same theorem gives

$$k^\star(256 \text{ KiB}, 2) = 2.$$

Thus two full-evidence probes with one non-remote probe form another feasible operating point. Coordination does not create capacity; it selects a particular point on the coverage–richness frontier.

a) Fixed-full overload validation.: For method C, Proposition 1 predicts per-probe queue growth over the 95 generation rounds. At low capacity,

$$Q_i(95) \approx 95 \left(B - \frac{B}{3} \right) = 7.9167 \text{ MiB},$$

while at mid capacity,

$$Q_i(95) \approx 95 \left(B - \frac{2B}{3} \right) = 3.9583 \text{ MiB}.$$

TABLE III
Formal methods and V1.3 terminology. Historical labels remain only for artifact traceability.

ID	V1.3 name	Historical artifact label	Runtime semantics
A	Fixed local	fixed-local	Always H-None; no application evidence upload.
B	Fixed summary	fixed-summary	Always H-Summary.
C	Fixed full	fixed-full	Always H-Full; no joint pre-admission.
D	I-QPG	independent-DPP	Per-probe queue-penalized selection; no joint admission.
E	C-QPG	central-DPP	Queue-penalized requests plus joint shared-capacity admission.
F	S-CQPG	shielded-DPP	Local safety filtering followed by the same joint admission.

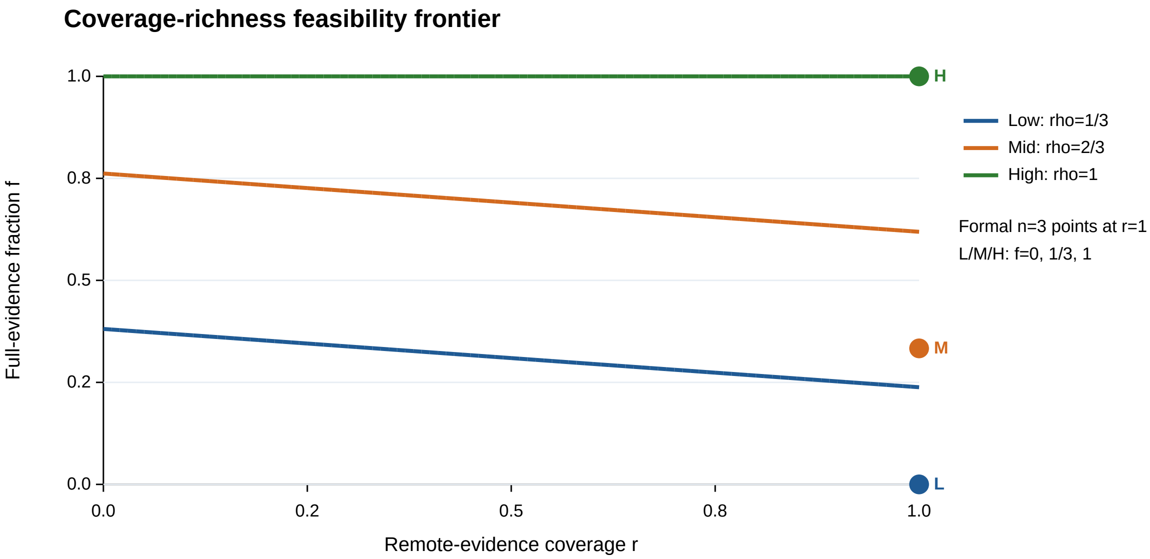

Fig. 1. Coverage–richness feasibility frontier. Curves are analytical; markers are the formal $n = 3$ coordinated operating points.

TABLE IV
Summary-preserving theoretical boundary and formal coordinated hardware operating points.

Capacity	$k^\star$	Predicted mix	E/F observed mix
Low	0	3 Summary	3 Summary
Mid	1	1 Full + 2 Summary	1 Full + 2 Summary
High	3	3 Full	3 Full

The sampled hardware maxima are approximately 7.9167 and 3.9584 MiB, respectively, closely matching the analytical overload law. At high capacity, $C = 3B$ and no deterministic overload remains.

The physical experiment therefore verifies both sides of the boundary: capacity-compliant coordinated tuples produce no service backlog, while the deliberately unadmitted fixed-full stream accumulates the predicted excess bytes.

B. RQ2: The Formal Independent Baseline Is a Synchronized Limit Cycle

At mid capacity, every formal I-QPG arm contains exactly 135 H-Full and 135 H-None measured actions. The core-required fraction is therefore exactly

$$\frac{135}{270} = \frac{1}{2},$$

with zero cross-session variation. Every one of the 135 core-required occurrences eventually returns, but none returns within the 1-s post-release SLO.

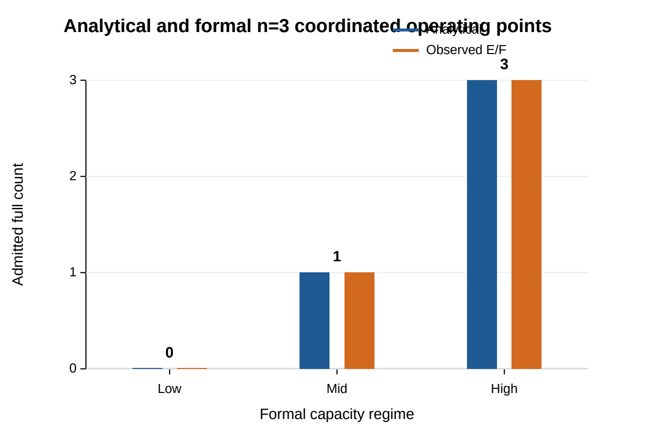

Fig. 2. Analytical and observed summary-preserving full-evidence counts.

This action pattern follows directly from Proposition 2. A synchronized H-Full round leaves approximately 43,691 B of sampled queue per probe, more than two orders of magnitude above the largest $v = 1$ positive-payload queue threshold of approximately 260.4 B. Hence all probes switch to H-None together. The following zero-payload round can drain the residual queue, restoring the state in which H-Full again has the highest utility.

The formal D result is therefore not evidence that independent control is intrinsically unable to use the shared service. It is evidence that a deterministic local controller can become phase-locked when identical probes observe symmetric queue states at synchronized epochs.

a) v sensitivity.: The methodology-repair replay first reproduced all 36 formal D session-capacity cells at $v = 1$ for nonzero action mix, application payload, core fraction, and sampled maximum queue. Across the sweep, the mid-capacity core-required fraction remains exactly 1/2 for every tested value through $v = 30,000$; it changes to 0.6222 only at $v = 100,000$. High capacity remains full-evidence for all tested values.

This result shows that the observed mid-capacity operating fraction is not a peculiarity of reporting only $v = 1$. It does not, however, imply that every action sequence is identical across v; low-capacity action compositions and detected queue cycles do change with the weight.

b) Desynchronization counterfactual.: The 20-seed software sensitivity changes only the probes' initial phases. No probe observes another probe's action or queue, and no joint admission is introduced. Under mid capacity, the long-run core-required fraction rises from 1/2 for the synchronized independent controller to exactly 2/3 for the desynchronized independent controller. The corresponding simulated on-time service coverage lies from 0.5000 to 0.5056 (mean 0.5028) across the fixed seeds, whereas the synchronized independent case is zero.

The result changes the interpretation of RQ2. The extreme formal failure is caused by synchronization rather than decentralization itself. C-QPG and S-CQPG instead choose a deterministic summary-preserving point with full fraction 1/3. The desynchronized independent controller chooses a richer full-evidence operating point but does not enforce the same per-round remote-evidence coverage policy.

Accordingly, V1.3 makes no claim that centralized admission universally outperforms a properly phased decentralized controller.

C. RQ3: Hardware Adds Timing Residuals Beyond Byte Feasibility

The analytical region predicts which payload combinations fit the application budget. It cannot predict scheduling jitter, OS delay, barrier waiting, Wi-Fi timing, serialization, or core-return latency. These effects become most visible when the byte constraint is non-binding.

At high capacity, C, D, E, and F all use H-Full for every measured event, so each core-capable method has identical evidence richness and core-required fraction. Mean post-release on-time core-result coverage is:

TABLE V
High-capacity physical post-release core-result coverage.

Method	Mean coverage
Fixed full (C)	0.9636
I-QPG (D)	0.9593
C-QPG (E)	0.9698
S-CQPG (F)	0.9651

The paired F–E mean difference is -0.00463 with descriptive 95% session-bootstrap interval $[-0.01327, 0.00278]$. The F–D difference is approximately $+0.00586$, and its interval also includes zero. Once the byte constraint is non-binding, the data provide no evidence of a coordinated-policy timing advantage.

Mid-capacity coordinated hardware results show the remaining timing effect more directly. E and F both admit exactly one full-evidence event per three generated events. Among the core-required subset, mean post-release on-time rates are approximately 0.8815 and 0.8935, respectively. Thus the ideal byte-feasible prediction of one full action does not imply perfect physical SLO attainment.

a) Release-time anchoring.: The formal experiment originally measured latency from actual event creation. The raw-timestamp reconstruction additionally anchors latency to the nominal scheduled release. Across the accepted formal data, the aggregate median release lateness is approximately 1972.34 ms for P1, 763.90 ms for P2, and 83.56 ms for P3. Across core-required cells, the difference between post-release and nominal-anchor on-time coverage ranges from 0 to 1.

Source and log inspection identify a mechanism for persistent phase offset: each probe creates an independent nominal clock before window 0, the core decision requires a three-probe barrier, unequal first-iteration waiting appears immediately, and the subsequent probe loop sleeps relative to the completed iteration rather than catching up to an absolute clock. This can preserve the initial offset.

The specific OS or network cause of the recurring P1>P2>P3 proposal arrival ordering is not identified by the available evidence and is therefore reported as unknown. In particular, the inverted ordering must not be interpreted as a Raspberry-Pi compute-speed comparison.

These findings also delimit the meaning of the primary SLO: the 1-s metric is a post-release service SLO, not a complete schedule-to-result response deadline.

D. RQ4: Scaling Conforms to the Integer Boundary and the Hardware Result Chain Is Reconstructable

For the summary-preserving case, define

$$f_n^\star(\rho) = \frac{1}{n}\left[\left\lfloor n\frac{\rho-\sigma}{1-\sigma}\right\rfloor\right]_0^n, \qquad \sigma = \frac{1}{8}.$$

The methodology-repair simulator evaluates $n \in \{3, 8, 16, 32\}$ and five normalized capacities. All 20 sim-

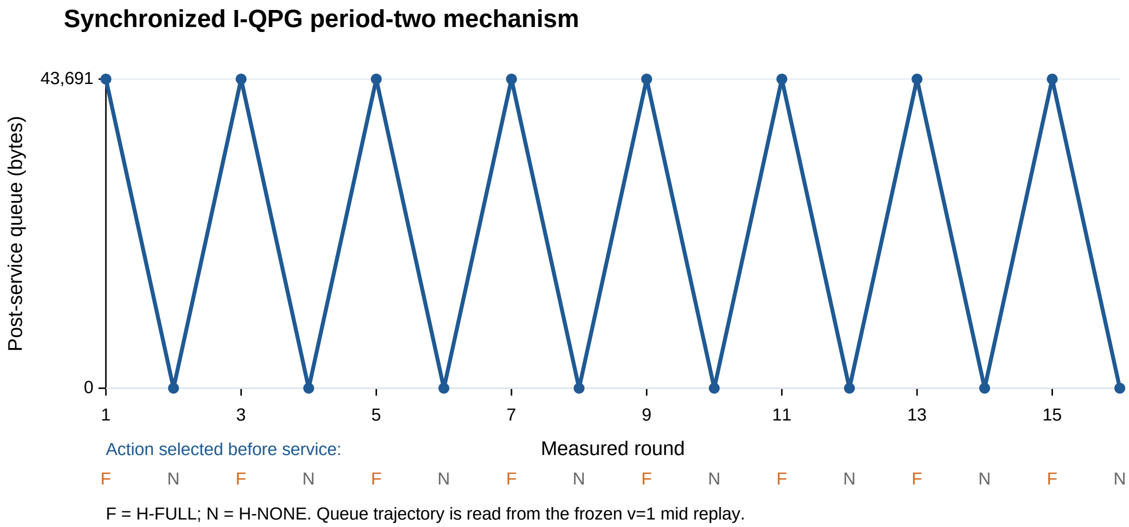

Fig. 3. Frozen $v = 1$ mid-capacity I-QPG queue trajectory and period-two action state.

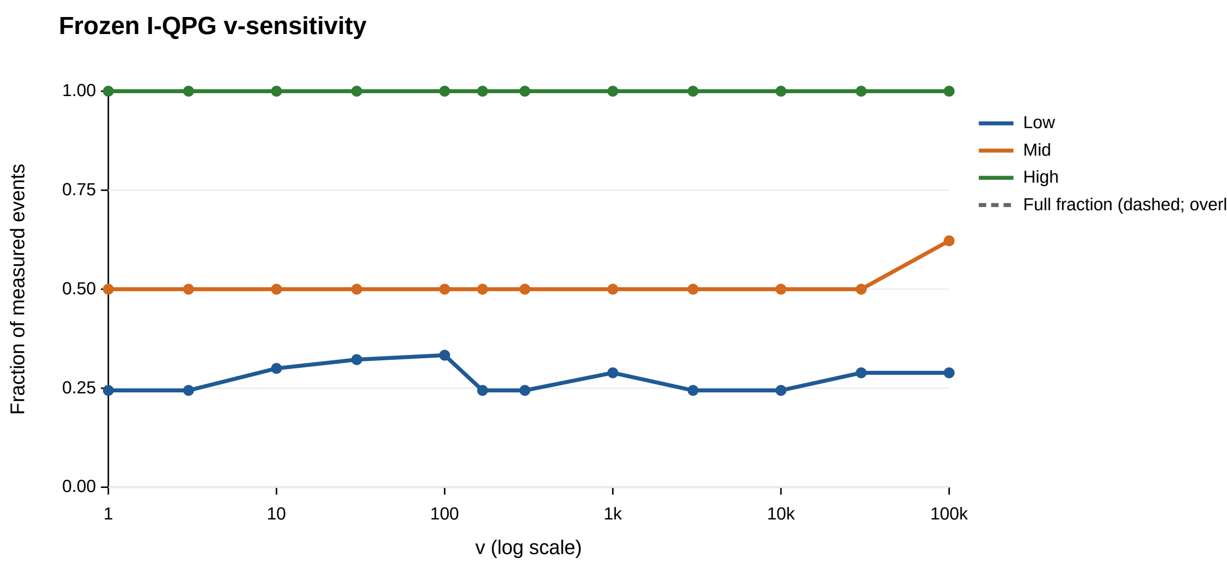

Fig. 4. Frozen-selector sensitivity across the 12 tested v values. Solid and dashed traces show core-required and full-evidence fractions; they overlap for this selector.

TABLE VI
Analytical maximum full count $k^\star$ for $m = n$; all 20 values match the discrete-event conformance simulation.

n	$\rho = 1/3$	$1/2$	$2/3$	$3/4$	1
3	0	1	1	2	3
8	1	3	4	5	8
16	3	6	9	11	16
32	7	13	19	22	32

ulated full counts match the analytical integer result exactly.

As n grows, integer granularity decreases and the boundary approaches

$$f^\star(\rho) = \frac{\rho - \sigma}{1 - \sigma}$$

inside the feasible interval. The simulation is therefore a conformance test of the closed-form boundary, not evidence of real 32-probe system performance.

a) Raw reconstruction.: The full-raw campaign audit independently rebuilds all 216 accepted arms from 58,320 measured occurrences. It reports zero raw-to-validator/event-metric mismatch and zero shared-cap violation across 63,720 audited service rounds. Rebuilt aggregate metrics, the public numeric candidate, and the

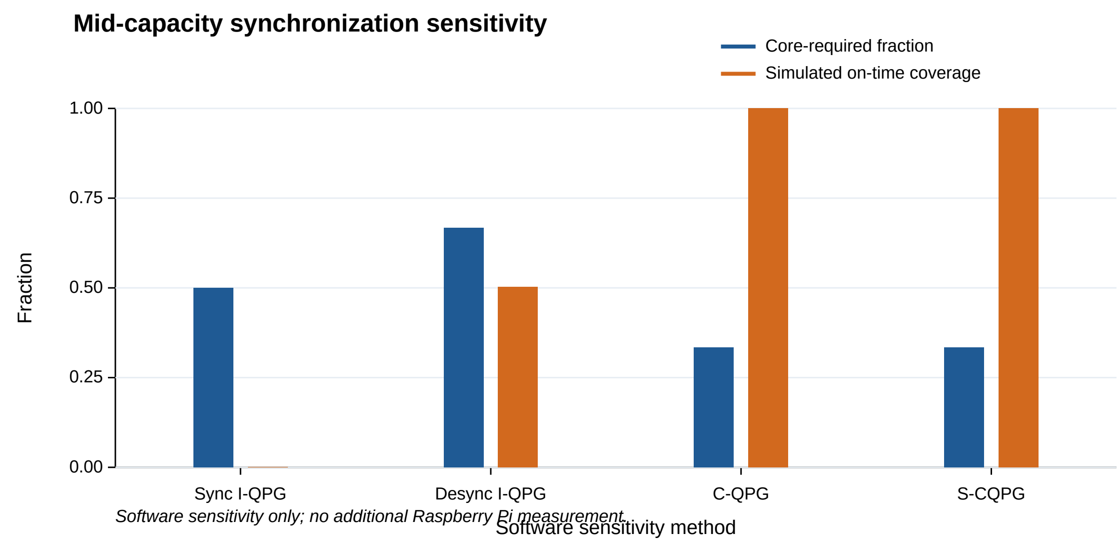

Fig. 5. Mid-capacity software sensitivity. Desynchronization changes only probe phases and introduces neither state exchange nor joint admission.

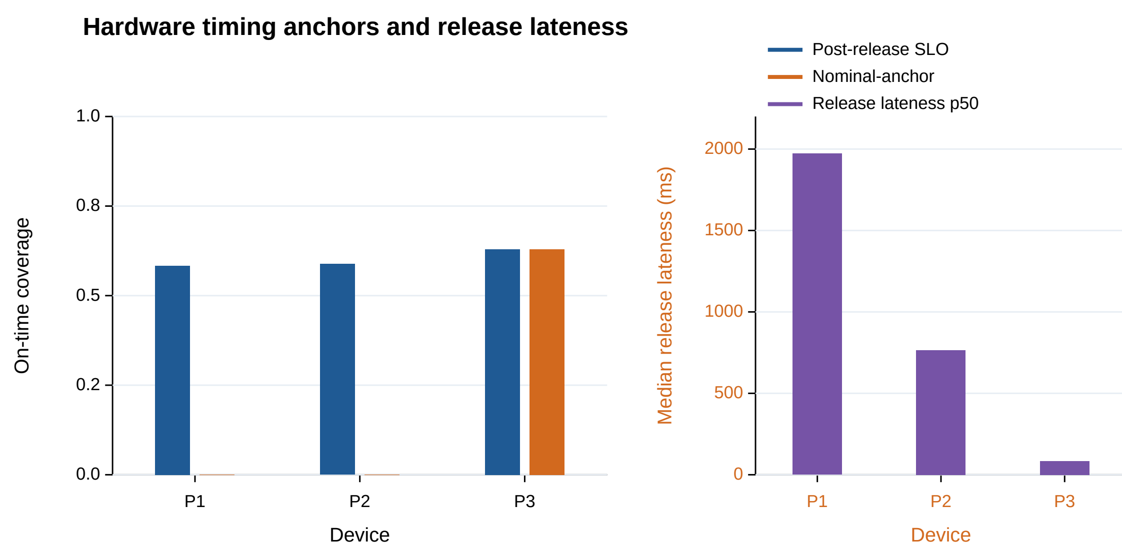

Fig. 6. Accepted-hardware timestamp reconstruction. On-time coverage is aggregated over core-required events; release lateness is shown separately.

accepted formal matrix are byte-identical to the delivered results.

The raw-audit verdict remains PASS_WITH_ISSUES because two archival gaps remain: the raw package lacks a direct S03-a01 SSD-failure incident artifact, and it does not contain the S07 pre-hardware launcher-abort files. Neither gap changes the reconstructed accepted arms, but both remain disclosed.

The separate V1.3 methodology-repair package has a PASS validation verdict: 432 v-sweep rows, 240 desynchronization rows, 648 timestamp-reconstruction rows, and 20 scaling rows are present with zero retained failure. Tasks A, B, and D are offline replay or simulation; only Task C is reconstructed from measured formal hardware timestamps.

E. Role of the Safety Filter

The safety-filtered method records 3,240 H-Full→H-Summary safety interventions under low capacity. Thus the filter is not a dormant code path. However, the unfiltered coordinated method reaches the same final low-capacity summary mix through shared-capacity admission.

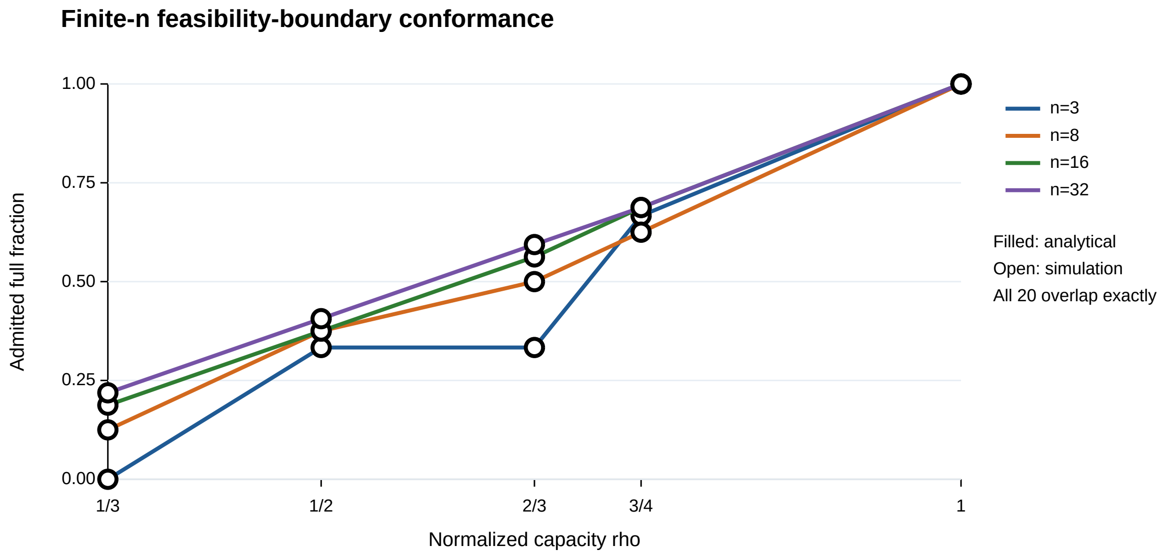

Fig. 7. Discrete-event conformance to the analytical finite-n boundary. Open simulated markers coincide with analytical points in all 20 cells.

At mid capacity, both E and F record 2,160 H-Full→H-Summary capacity-admission degradations, with no additional safety-filter action change. At high capacity neither mechanism degrades full evidence.

The evidence therefore supports a narrow claim: the filter provides an explicit and auditable location for local feasibility predicates. The present hardware campaign does not demonstrate that this layer improves end-to-end performance beyond coordinated capacity admission.

F. Classifier Correctness Is a Controlled Diagnostic

The formal feature replay yields perfect classification on the measured occurrences for all frozen detector paths used in the event ledger. Consequently, the old "on-time-and-correct" field is numerically identical to on-time service whenever core execution is required.

This ceiling is not interpreted as a classifier result. Independent detector audit shows that the complete frozen test set is itself nearly perfect for the heavy and medium models, while most validation error is associated with Recon-PortScan, which is absent from the formal replay. Exact feature-vector overlap also exists across frozen data partitions.

For these reasons, V1.3 treats detector correctness as a controlled condition and does not generalize the formal F1 values to open-world IoT intrusion detection.

VI. Discussion

A. The Main Result Is a Frontier, Not a Ranking

The methodology repair changes the interpretation of the entire experiment. The original method ranking implicitly treated more timely full-core results as evidence that coordination was superior. The desynchronization counterfactual shows why that interpretation is too strong.

At mid capacity, a coordinated policy that preserves remote evidence for all three probes occupies the point

$$(r, f) = \left(1, \frac{1}{3}\right).$$

An independent controller can instead occupy a point with a larger long-run full fraction by allowing less uniform evidence coverage over time. Neither point dominates without assigning value to remote coverage versus full-evidence richness.

The contribution of coordinated admission is therefore deterministic operating-point enforcement. It can guarantee that the chosen full/summary tuple lies inside the configured capacity region. It does not imply that the selected point maximizes every downstream security objective.

B. Why the Hardware Experiment Is Still Necessary

Several headline byte quantities are analytically predictable. This is a strength rather than a reason to discard the hardware campaign: a correct network model should predict deterministic byte behavior.

The hardware data answer the part the byte model cannot: how close a feasible tuple comes to a real response SLO. Even when high capacity makes all three full requests byte-feasible, approximately 3–4% of core returns can still fall outside the 1-s post-release SLO. Likewise, the nominal-clock reconstruction reveals orchestration delays that are invisible if latency begins only when the event is eventually released.

The role of the 216-arm campaign is thus validation and residual measurement, not discovery of an arithmetic capacity law.

C. Synchronization Is a Systems Variable

The formal I-QPG experiment illustrates an often-overlooked property of distributed control. Identical local policies do not imply independent behavior when their observations and decision epochs are synchronized. The shared service couples their queues; identical queues then feed back into identical decisions.

The 20-seed phase sensitivity demonstrates that this coupling can be changed without exchanging any controller state. This motivates treating probe phase, jitter, or randomized tie-breaking as an explicit systems design parameter in decentralized evidence scheduling.

D. Evidence Richness Is Not Security Utility

A summary and a full-evidence upload provide different downstream information. Reducing application bytes can improve timeliness while also reducing forensic richness. Conversely, sending more full evidence can improve the material available to a core but reduce the fraction that meets its response horizon.

Accordingly, payload reduction alone is not a security improvement. The analytical frontier deliberately exposes both dimensions rather than collapsing them into a single "bandwidth saving" score.

VII. Limitations

Three-probe physical scale. The formal hardware testbed uses three probes. The multi-n study is a discrete-event conformance test of the analytical boundary and must not be interpreted as physical performance validation at 8, 16, or 32 nodes.

Desynchronization is a software sensitivity. The phase-randomized independent baseline is not a new Pi experiment. It establishes that the synchronized failure is not intrinsic to decentralization, but it does not measure real Wi-Fi timing for a desynchronized physical deployment.

Application budget versus physical link rate. Low/mid/high are application-byte service budgets per coordination round, not measured PHY throughput.

Timing anchor. The primary 1-s metric starts at actual event creation. The nominal-anchor analysis demonstrates that schedule-inclusive results can be materially worse. Both anchors are therefore reported separately.

Replay scope. The workload is a frozen CICIoT2023-derived feature replay rather than an uncontrolled packet-level deployment. Recon-PortScan, a major validation error family, is absent from the formal replay, and exact feature-vector repetition exists across frozen data splits.

Shield separation. The tested campaign does not contain a case where a local device constraint rejects an action that is independently feasible under the shared network budget. The safety filter is therefore an implementation interface and audit mechanism in this paper, not an independently proven performance layer.

Evidence value. The experiment measures evidence coverage, richness, byte demand, and timeliness, but not the downstream incident-resolution value of a summary versus full evidence. Future work should connect the frontier to measured forensic or cross-probe correlation utility.

VIII. Reproducibility and Provenance

The formal experiment uses protocol NSF12_20260906_RC1. The accepted scientific attempts are S01-a01, S02-a01, S03-a02, S04-a02, S05-a01, S06-a01, S07-a02, and S08–S12-a01. S03-a01, S04-a01, and S07-a01 are invalid infrastructure attempts and are excluded in full rather than replaced arm-by-arm.

The compact audit package, formal campaign package, full-raw audit results, and methodology-repair package are separately hashed. The V1.3 methodology repair did not modify the formal V1.2 raw evidence, runtime source, Raspberry Pis, manuscript, or repository state.

Before public release, the final artifact should expose persistent checksums and a durable archival identifier. Dataset redistribution rights should be checked separately; the paper need not redistribute third-party CICIoT2023 content to make the experiment code and derived result chain reproducible.

IX. Conclusion

This paper reframes multi-probe IoT edge intrusion detection as an evidence-service allocation problem over a shared bottleneck.

The analytical result is simple but consequential: when m probes retain remote evidence and k of them request full evidence, one-round byte feasibility is determined by

$$kB + (m-k)S \le C.$$

Its normalized form,

$$\sigma r + (1-\sigma) f \le \rho,$$

makes explicit the trade-off between remote-evidence coverage and full-evidence richness.

The three-probe hardware campaign validates this boundary at its formal low, mid, and high operating points and verifies the associated fixed-full overload law. It also reveals behavior that a static capacity equation cannot predict. The frozen independent queue-penalized controller enters a synchronized full/none period-two orbit at mid capacity; parameter sensitivity shows that the operating fraction persists across a wide range of queue weights, while phase randomization recovers substantially more independent full-evidence service without central admission. The correct conclusion is therefore not that centralized control universally outperforms decentralization, but that explicit admission can deterministically enforce a chosen point on the coverage–richness frontier.

Finally, physical timestamps show why byte feasibility is only the first layer of the problem. Even feasible full-evidence tuples experience real execution and wireless timing variation, and schedule-relative release offsets can materially change an end-to-end SLO interpretation.

Across the 216-arm formal campaign, the event-level result chain can be reconstructed from raw evidence with no accepted-arm metric mismatch or shared-cap violation. The resulting contribution is a narrower but stronger networking result: NetShield provides a framework for reasoning about, enforcing, and measuring evidence coverage, richness, queueing, and service timeliness under a shared edge bottleneck.

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